

Raspberry Pi: A Step-by Step Guide

Introduction:

Setting up your Raspberry Pi involves a series of straightforward steps to establish both hardware and software connections, ensuring a seamless and efficient user experience. This guide provides a step-by-step walkthrough from connecting the micro-HDMI to HDMI cable to enabling SSH (Secure Shell) for remote access. Follow the instructions below to get your Raspberry Pi up and running for various applications.

1. Hardware Setup:

- Connect the Micro-HDMI to HDMI cable, linking your Raspberry Pi to a HDMI monitor.
- Plug in the Pi hub for keyboard and mouse connectivity.
- Power on the Raspberry Pi using either a USB-C power with a wall adapter or by plugging in a battery into the drone.

2. Initial Login and User Setup:

- Log in and follow the commands for creating a username and password (ensure to keep them secure as they will be needed later).

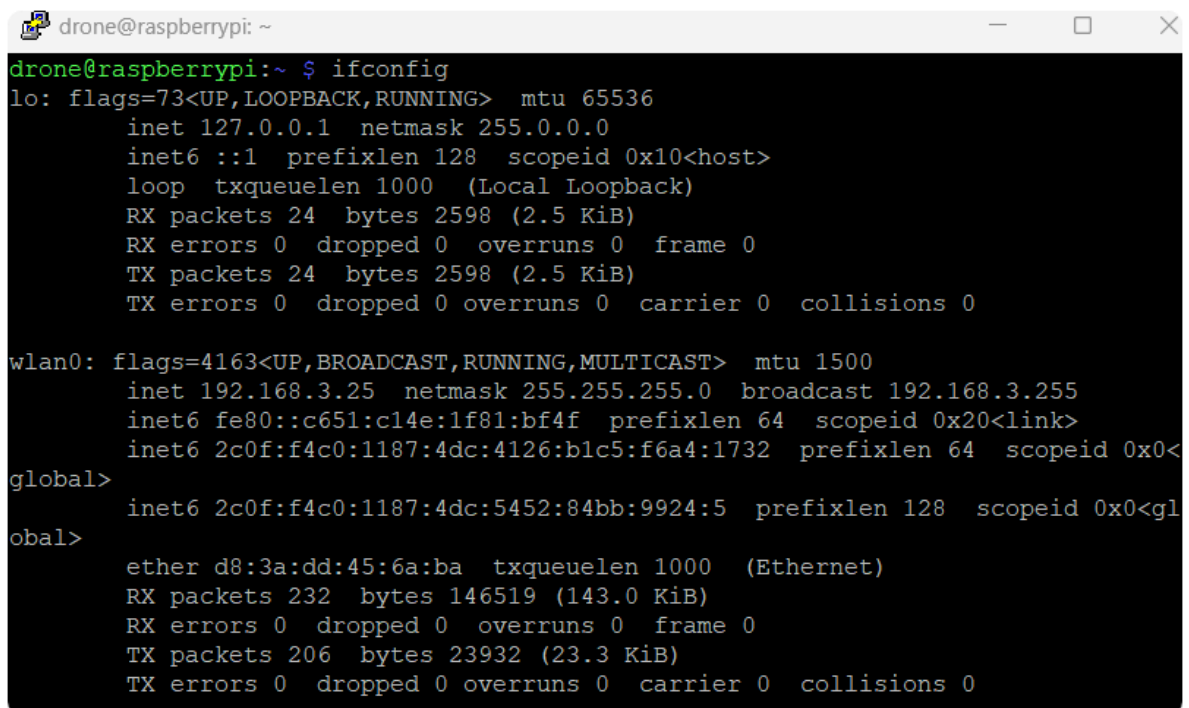
3. Wi-Fi Configuration:

- Select and connect to your Wi-Fi network, providing the Pi with internet access.

4. Identifying Hostname and IP: • After the Pi reboots and logs in, go to the terminal and type "hostname" (note it down).

```
drone@raspberrypi:~ $ hostname
raspberrypi
drone@raspberrypi:~ $
```

- Type "ifconfig" in the terminal and note down the IP under wlan0.



```
drone@raspberrypi: ~
drone@raspberrypi:~ $ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 24 bytes 2598 (2.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 24 bytes 2598 (2.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

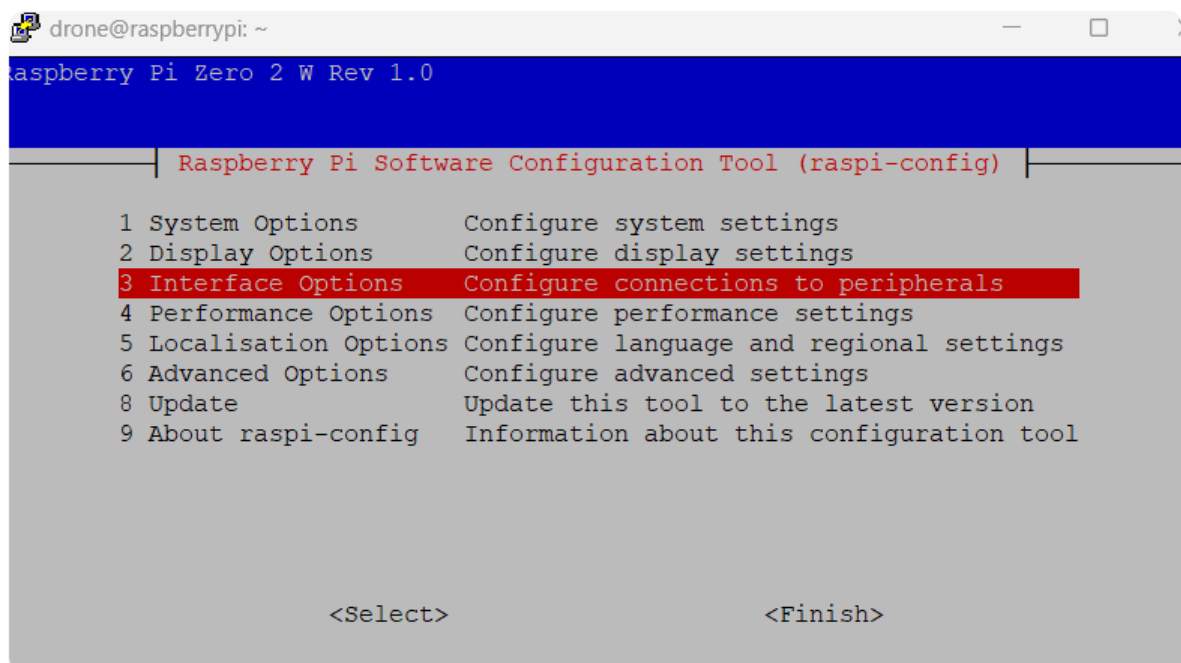
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.3.25 netmask 255.255.255.0 broadcast 192.168.3.255
    inet6 fe80::c651:c14e:1f81:bf4f prefixlen 64 scopeid 0x20<link>
    inet6 2c0f:f4c0:1187:4dc:4126:b1c5:f6a4:1732 prefixlen 64 scopeid 0x0<
global>
    inet6 2c0f:f4c0:1187:4dc:5452:84bb:9924:5 prefixlen 128 scopeid 0x0<gl
obal>
    ether d8:3a:dd:45:6a:ba txqueuelen 1000 (Ethernet)
    RX packets 232 bytes 146519 (143.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 206 bytes 23932 (23.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

5. Enabling SSH for Remote Access:

Open the terminal and type "sudo raspi-config."

```
drone@raspberrypi:~ $ sudo raspi-config
```

- Navigate to interface options.



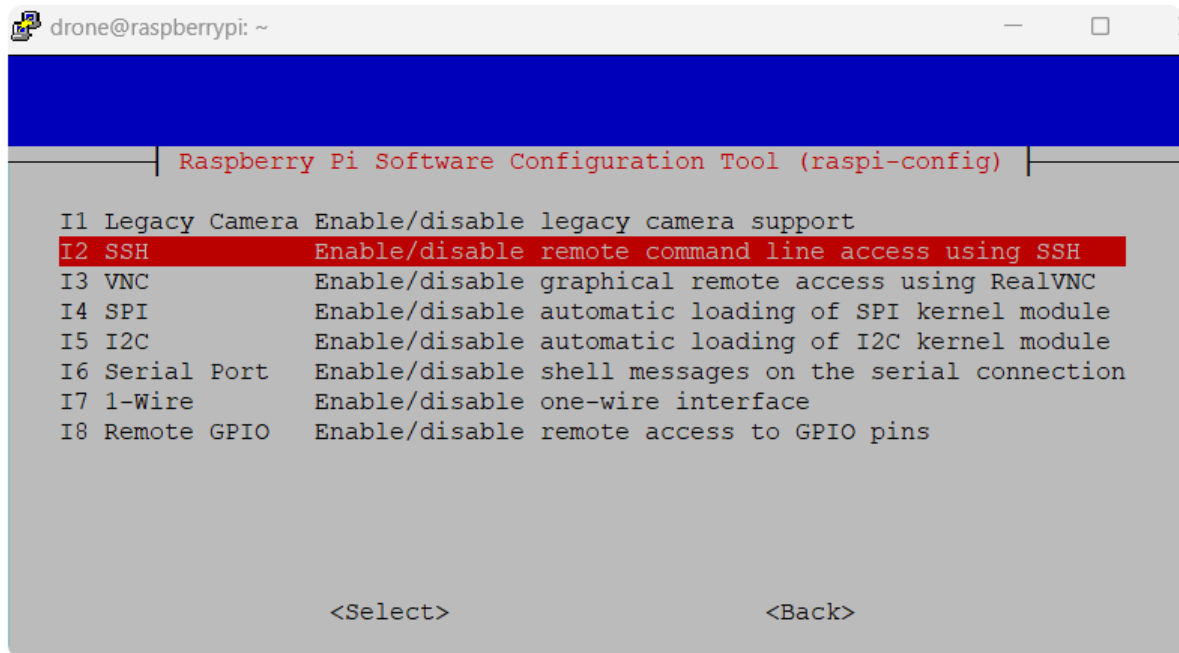
```
drone@raspberrypi: ~
raspberrypi Pi Zero 2 W Rev 1.0

Raspberry Pi Software Configuration Tool (raspi-config)

1 System Options      Configure system settings
2 Display Options     Configure display settings
3 Interface Options   Configure connections to peripherals
4 Performance Options Configure performance settings
5 Localisation Options Configure language and regional settings
6 Advanced Options    Configure advanced settings
8 Update              Update this tool to the latest version
9 About raspi-config  Information about this configuration tool

<Select>              <Finish>
```

- Select SSH



- Enable SSH by selecting "yes" and confirming with OK, then finish.

